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ORIGINAL ARTICLE

Chronic pain, psychological distress, and quality of life in males with Duchenne muscular dystrophy

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Abstract

Aim: To describe chronic pain in Duchenne muscular dystrophy (DMD) from children's/adolescents' perspectives, explore patient variables associated with self-reported pain, and examine the relationship between chronic pain, psychological functioning, and health-related quality of life (HRQoL).

Method: This observational study included a paediatric cohort (aged 8–18 years; median age 9 years 4 months) with DMD under multidisciplinary care (n = 45). Clinical data of the latest visit were extracted from the electronic health record and assessment of pain, psychological distress, and HRQoL were performed during the same visit.

Results: Thirty-two patients reported pain during the previous 4 weeks, and 18 reported persistent or recurrent chronic pain. Average pain intensity of chronic pain was mild, with regions of the legs (n = 12), lower back (n = 6), hips (n = 6), and shoulder (n = 6) most frequently affected. Older age, higher body mass index, being non-ambulatory, wheelchair dependency, and spinal deformities were contextual variables related to the presence of chronic pain. Furthermore, chronic pain was significantly associated with psychological distress and reduced HRQoL in paediatric patients with DMD.

Interpretation: Chronic pain in paediatric DMD is frequent and widespread, highlighting the need for pain to be addressed in the routine care of affected young people. Chronic pain may make a significant contribution to psychological distress and impaired HRQoL in paediatric patients with DMD.

Duchenne muscular dystrophy (DMD) is the most common form of childhood-inherited neuromuscular disorder, with an incidence of 1 in 3500 to 6000 live male births.[1] DMD is an X-linked recessive disease caused by mutations in the dystrophin gene, resulting in progressive muscle degeneration and loss of ambulation.[2] Additional complications of the musculoskeletal, respiratory, and cardiac systems develop as DMD progresses.[3] Over recent decades, improved management with corticosteroid therapy and non-invasive ventilation has led to better survival.[4] Currently, there is no cure for DMD; the aim of clinical interventions is to delay disease progression and improve health-related quality of life (HRQoL).[3]

Chronic pain is a common problem among people with muscular dystrophies and has emerged as a significant risk factor for poor HRQoL in these populations.[5–7] The aetiology and phenotypes of muscular dystrophy-related chronic pain are largely unknown. The localization and source of chronic pain reported by patients mainly point towards a musculoskeletal origin of pain.[8–10] For people diagnosed with neuromuscular disorders, pain may transition from acute to chronic, yielding additional challenges for both

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Abbreviations: DMD, Duchenne muscular dystrophy; HRQoL, health-related quality of life; PedsQL 4.0, Pediatric Quality of Life Inventory version 4.0.

patients and care providers.[8] Our previous meta-analysis has demonstrated that chronic pain affects a high proportion (60–68%) of people across a range of muscular dystrophies, including adult and paediatric populations, with many describing their pain as frequent, widespread, and severe.[5]

Existing chronic pain data in muscular dystrophies come primarily from adult cohorts that focus on two main subtypes: facioscapulohumeral muscular dystrophy and myotonic dystrophy. Knowledge of mechanisms, prevention, and management of chronic pain in other muscular dystrophies seems to be limited, especially in the paediatric population. Some studies have investigated pain in paediatric patients with DMD.[7,11,12] These studies, however, did not focus on 'chronic' pain based on an accepted definition; instead, they explored pain in general, including acute pain that may have been transient in nature. To our knowledge, no study has further explored the impact of chronic pain on psychological functioning and HRQoL in DMD. Identifying chronic pain, understanding the underlying determinants, and justifying its contribution to psychological distress and poor HRQoL may help develop therapeutic strategies to manage chronic pain and improve HRQoL. Furthermore, early preventive care and active pain management in childhood may lead to better intervention outcomes and reduce a child's vulnerability to pain later in life.

To promote best clinical practice in the prevention and management of chronic pain in paediatric patients with DMD, an insightful comprehension of patients' perspectives of pain is needed. Considering the nature of pain (a subjective and individual experience), self-report measures appropriate to the child's age, language, and cognitive maturity are recommended.[13] Studies have shown that the actual agreement between parent and child report on pain symptoms seems to be poor to fair in DMD.[12] Proxy reports may lead to an underestimation of the child's symptoms. Therefore, this cross-sectional cohort study aimed to describe chronic pain in DMD from children's/adolescents' perspectives, explore patient variables associated with self-reported pain, and examine the relationship between chronic pain, psychological functioning, and HRQoL.

What this paper adds

- Chronic pain differs in aetiology, scope, and nature compared with acute pain in paediatric Duchenne muscular dystrophy (DMD).
- Older age, higher body mass index, being non-ambulatory, wheelchair dependency, and spinal deformities are important patient variables.
- Chronic pain is significantly associated with psychological distress and reduced health-related quality of life in paediatric DMD.

METHOD

This was a cross-sectional cohort study based on both primary and secondary data extracted from patients' hospital records.

Patients and setting

A convenience sample of young males with genetically confirmed DMD was recruited between January 2020 and July 2021 through a large public children's hospital (Shenzhen Children's Hospital) covering patients from southern China (including Guangdong, Guangxi, Hunan, Hubei, Jiangxi, Fujian, and Guizhou provinces). Paediatric patients with DMD were included if they were (1) aged 8 to 18 years and (2) under multidisciplinary care with routine follow-up visits. Patients who had cognitive impairments or speech and language dysfunction that hindered their comprehension of pain questions or self-expression of pain complaints were excluded. The design of this study gained ethical approval from the Ethics Committee for Medical Research of Shenzhen Children's Hospital. Patients and their caregivers both signed the informed consent.

Clinical records

The following clinical information of the latest visit was reviewed for each patient, where available. (1) Clinic reports from multiple departments the patient attended, including neurological, respiratory, cardiology, gastroenterology, orthopaedic, radiological, and surgical clinics. (2) Serial reports of physiotherapy assessments, which included muscle strength, functional scales (the Vignos and Brook grades), standardized motor function assessment (the Motor Function Measure for neuromuscular disorders), and timed function tests (10 m walk/run, climbing four stairs). (3) Results of genetic testing. The most recent value was used when data were available at more than one timepoint for the same patient.

In addition to the above items, several patient-reported outcome measures on pain, psychological distress, and HRQoL were added after consultation with the patient and parents about their willingness to participate in the study. The children who agreed to participate completed these measures during the visit to their regular physiotherapist at the rehabilitation centre. All participants filled in the questionnaires under the guidance of an investigator (M.H.), providing the opportunity to clarify questions and explore more details about the child's responses as needed.

Pain assessment

Chronic pain in this study was defined as persistent or recurrent pain lasting for 3 months or longer.[14] In addition to standardized patient-reported outcome measures, the questionnaire included questions pertaining to pain was

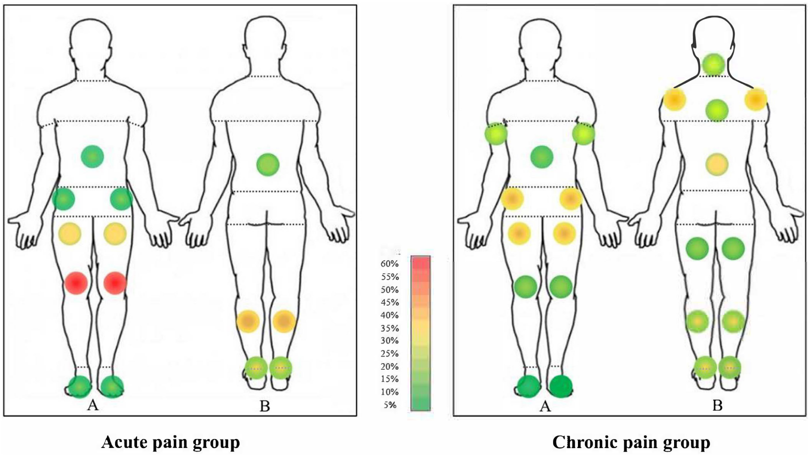

FIGURE 1 Pain frequency using localized body maps for acute and chronic pain groups. The colour grid indicates the percentage of participants within the group who reported pain in that specific area. A, anterior; B, posterior

based on literature exploring pain in other muscular dystrophies[9,15] and the Initiative on Methods, Measurement, and Pain Assessment in Clinical Trials guideline.[16] The following question was asked to recognize the presence of pain and chronic pain: Have you experienced any pain in the past 4 weeks other than occasional pain such as headache or toothache? If yes, had the pain started more than 3 months previously? Patients who reported pain were asked to respond to further questions about their pain, including pain course, frequency and duration of episodes, pain location, pain intensity, pain interference with daily functioning, pain control measures, history of clinical pain assessment and management, and self-perceived causes of pain. A pilot study was performed in 10 typically developing male children aged 8 to 12 years with acute or chronic pain. The questionnaire pertaining to pain was tested among these children to ensure that all questions were clear and understandable to the age group. Modifications were made if more than 20% of the children found the question difficult to answer.

Pain location

In the questionnaire, patients were asked to mark all pain they had experienced in the previous 4 weeks on a body map. The pain sites were then categorized into eight body regions (as presented in Figure 1): head, shoulders, arms and hands, abdomen, lower back, hips, legs, and feet. The number of different pain sites was calculated to reflect pain extent.

Pain intensity and severity

The average and worst pain intensities of all pain in general during the previous week were estimated by the Numeric Pain Rating Scale.[17] The child was asked to select a whole number (0–10 integers) that best reflected the intensity of their pain, with '0' suggesting 'no pain' and '10' suggesting 'pain as bad as you can imagine'. The Numeric Pain Rating Scale has been found to be a valid and reliable self-report measure of pain intensity for children from 8 years of age.[18] Cut-off points for pain severity are decided in accordance with the International Association for the Study of Pain,[14] where Numeric Pain Rating Scale scores of 0 to 3 suggest mild pain, scores between 4 and 6 suggest moderate pain, and scores of 7 and higher suggest severe pain.

Pain interference

The impact of pain on daily functioning during the previous week was evaluated by the modified version of the Pain Interference Scale from the Brief Pain Inventory.[19] To make the 7-item Pain Interference Scale (general activity, mood, mobility, schoolwork, relationships with others, sleep, and enjoyment of life) more valid for people with disabilities, we replaced the item 'walking ability' by 'mobility: ability to get around'. In addition, the item 'normal work' was changed to 'school work' to better suit children and teenagers aged 8 to 18 years.

Pain coping and communication

Patients were also asked specific questions to indicate their pain control measures (e.g. What did you usually do to reduce your pain? Did you usually take analgesics when you experienced pain?), whether they had communicated about their pain with parents or clinicians, and whether they had received clinical pain assessment and management from health professionals.

Self-perceived cause of pain

Self-perceived cause of pain was assessed by an open question: What do you perceive to be the cause of your pain? If the patients had difficulties understanding the question, further explanations were provided by the investigator. Some other forms of questions helped the child to understand what we were asking including: What do you think is the reason for your pain? In what situations do you feel pain?

Psychological distress and HRQoL

Anxiety and depression symptoms were measured using the Hospital Anxiety and Depression Scale.[20] This consists of 14 items: seven of the items relate to anxiety and seven relate to depression. Each item is scored from 0 to 3. Total scores were calculated separately for anxiety and depression. Ratings of 8 or higher are generally considered as cut-off scores for anxiety and depression, with ratings of 8 to 10 indicating mild anxiety/depression, 11 to 14 indicating moderate anxiety/depression, and 15 to 21 indicating severe anxiety/depression. The Hospital Anxiety and Depression Scale has been validated for its use in paediatric patients with chronic health conditions.[21]

HRQoL, which reflects a patient's perceptions of the impact of an illness and its treatment on their own functioning and well-being, was assessed using the Pediatric Quality of Life Inventory version 4.0 (PedsQL 4.0).[22] The PedsQL 4.0 is reliable and valid in children diagnosed with chronic disease.[23] It comprises 23 items that assess four domains of health: physical functioning, emotional functioning, social functioning, and school functioning. For each question, respondents are asked to indicate to what degree a statement has been true during the previous month (five response categories, ranging from 'never' to 'almost always'). These responses are reverse scored and linearly transformed to a 0 to 100 scale, with a higher score indicating a higher HRQoL. Patients were asked to complete age-appropriate self-report versions of the PedsQL 4.0. The patients were asked to complete the PedsQL on their own without any suggestions; support and guidance from parents and investigators were only allowed to assist the children to understand the questions.

Data analysis

All analysis used SPSS version 26 (IBM Corp., Armonk, NY, USA). A Shapiro–Wilk test was used to inspect the normality of all the data. Data are presented as the mean (standard deviation [SD]) for normally distributed data or median (interquartile range [IQR]) for non-normally distributed data for continuous and ordinal variables. Other categorical variables are shown with frequency and percentage. Group comparisons were conducted using an independent samples t-test or one-way analysis of variance for normally distributed data (Welch's t-test or analysis of variance were used in the case of unequal variances between groups) and the Kruskal–Wallis test or Mann–Whitney U test for non-normally distributed continuous and ordinal data. For nominal data, group comparisons were performed using a χ^2 test or Fisher's exact test, as appropriate. For multiple comparisons, p-values were adjusted with Bonferroni correction.

Spearman's rank correlation analysis was used to assess the relationship between patient variables and chronic pain, with 0.10 to 0.39 considered weak, 0.40 to 0.69 considered moderate, and 0.70+ considered strong.[24] To explore the association between chronic pain and psychological functioning and quality of life, multiple linear regression models were performed to calculate β estimates and 95% confidence intervals (CIs). The β coefficient reflects the degree of change in the outcome variable for every 1 unit of change in the predictor variable. Four independent variables of interest were entered in a linear regression model based on the sample size.[25] For the association between chronic pain and domains of HRQoL, only age, body mass index (BMI), and ambulation status were adjusted. Psychological variables were excluded from the model because of potential mediational effects. For the association between chronic pain and psychological functioning, age, BMI, and ambulation status were also adjusted according to the results of univariate analyses. Residual plots and R^2 were used to evaluate the model performance. The R^2 statistic indicates the percentage of the variance in the dependent variable that the independent variables explain collectively. Differences were considered significant at $p < 0.05$, and all statistical tests were two-tailed. Occasional missing data were replaced by imputed values generated using a multiple imputation method. Where there was a failure to respond to an item by more than 30%, the item was omitted.[11] The sample size of the study was decided according to the rule of thumb suggested by Harrell[25] that 10 participants per variable was the minimum required sample size for linear regression models to ensure accurate prediction.

RESULTS

Demographic and clinical characteristics

Demographic and clinical data are presented in Table 1. Fifty-four eligible patients with DMD were recruited, but nine were excluded because of their unwillingness to participate in the study (n = 4) or incompetency in pain communication owing to cognitive impairments (n = 2) or language/speech problems (n = 3). Finally, 45 were included in this study. The median age of the entire cohort was 9 years 4 months (range 8 years – 15 years 5 months). Most patients received regular corticosteroid treatment under a regime of daily prednisone (n = 39). Most were independent ambulators (n = 40) with mild to moderate locomotor impairments and three of them were only able to walk indoors but used a wheelchair outdoors. Five patients were permanent wheelchair users. Twelve patients were diagnosed with mild scoliosis through X-ray imaging. Twelve patients experienced mild obstructive sleep apnoea syndrome, of whom nine depended on night-time

TABLE 1 Demographic and clinical characteristics of patients with Duchenne muscular dystrophy by pain group

Characteristic	Total	Pain group: No pain	Acute pain	Chronic pain	*p*
n (%)	45 (100)	13 (29)	14 (31)	18 (40)	
Demographics					
Age, median (IQR), years	9 (8–10)	8 (8–9)	9 (9–11)	10 (9–12)	**0.004**[a]
Sex, male, *n* (%)	45 (100)	13 (100)	14 (100)	18 (100)	
School education, *n* (%)					
Regular school	44 (98)	13 (100)	14 (100)	17 (94)	
Special school	0	0	0	0	
Home learning	1 (2)	0	0	1 (6)	
Primary caregivers, *n* (%)					0.927
Parents	40 (89)	12 (92)	12 (86)	16 (89)	
Father or mother	1 (2)	0	0	1 (6)	
Grandparents	4 (9)	1 (8)	2 (14)	1 (6)	
Educational level, *n* (%)					0.900
Primary	7 (16)	1 (8)	2 (14)	4 (22)	
Lower secondary	12 (27)	4 (31)	4 (29)	4 (22)	
Higher secondary	10 (22)	2 (15)	4 (29)	4 (22)	
Tertiary/university	16 (36)	6 (46)	4 (29)	6 (33)	
Clinical data					
Age at diagnosis, median (IQR), years	4 (3–6)	3 (2–4)	4 (3–7)	4 (4–6)	0.115
Use of steroids, *n* (%)					0.406
Never	6 (13)	2 (15)	3 (21)	1 (6)	
Past	0	0	0	0	
Current	39 (87)	11 (85)	11 (79)	17 (94)	
Duration of use, median (IQR), months	35 (12–48)	14 (2–34)	24 (6–39)	48 (33–60)	**0.003**[a]
Regimen, *n* (%)[b]					
Prednisone, daily	39 (100)	11 (100)	11 (100)	17 (100)	
Body composition, mean (SD)					
Stature, m	1.31 (0.12)	1.27 (0.10)	1.34 (0.13)	1.31 (0.13)	0.336
Body mass, kg	31.13 (7.94)	27.03 (5.28)	31.73 (5.92)	33.62 (9.85)	0.066
BMI, kg/m^2	17.94 (2.45)	16.57 (1.85)	17.57 (2.21)	19.21 (2.49)	**0.007**[a]
BFP	45.53 (21.18)	36.46 (16.47)	50.50 (19.30)	48.22 (24.38)	0.181
BMD	–1.55 (0.76)	–1.42 (0.70)	–1.63 (0.74)	–1.59 (0.86)	0.772
Surgical history, *n* (%)	3 (7)	1 (8)	0	2 (11)	0.618
Scoliosis, *n* (%), mild	12 (27)	1 (8)	3 (21)	8 (44)	0.064
Use of mechanical ventilation, *n* (%), night-time NIV	9 (20)	0	3 (21)	6 (33)	0.077
FVC (% predicted), median (IQR)	95.6 (91.3–98.3)	95.7 (91.6–98.3)	93.3 (91.0–97.7)	95.7 (89.0–99.6)	0.852
Cardiac function					
LVEF, median (IQR), %	64.0 (62.6–65.8)	64.4 (62.3–66.2)	65.3 (63.2–66.5)	63.1 (62.4–64.5)	0.176
Sleep apnoea, *n* (%)					
Mild OSAS	12 (27)	0	5 (36)	7 (39)	**0.027**[a]
Physical function					
Ambulation, *n* (%)					**0.017**[a,c]
Ambulant	40 (89)	12 (100)	14 (100)	13 (72)	

TABLE 1 (Continued)

		Pain group			
Characteristic	**Total**	**No pain**	**Acute pain**	**Chronic pain**	***p***
Non-ambulant	5 (11)	0	0	5 (28)	
Use of wheelchair, *n* (%)					**0.019**[a,c]
Never	37 (82)	13 (100)	12 (86)	12 (67)	
Intermittent	3 (7)	0	2 (14)	1 (6)	
Permanent	5 (11)	0	0	5 (28)	
Orthotics, *n* (%)	38 (84)	10 (71)	12 (85)	16 (89)	0.906
MRC%, median (IQR)	80.0 (70.0–90.0)	87.9 (81.1–90.4)	76.8 (69.8–90.2)	80.0 (60.0–85.2)	0.054
Vignos grade, median (IQR)	1.0 (1.0–2.0)	1.0 (1.0–1.0)	1.0 (1.0–2.3)	1.0 (1.0–9.0)	**0.003**[a,c]
Brook grade, median (IQR)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	1.0 (1.0–1.0)	1.0 (1.0–2.0)	0.230
MFM, median (IQR)					
Total score	85.4 (75.5–88.5)	87.5 (84.7–92.7)	78.1 (72.6–87.5)	83.8 (56.2–89.0)	**0.043**[a,d]
D1	64.1 (41.1–79.5)	76.9 (61.5–88.4)	50.0 (37.8–75.0)	61.5 (7.7–79.5)	**0.030**[a,d]
D2	97.2 (94.4–100.0)	97.2 (95.8–97.2)	97.2 (94.4–100.0)	97.2 (88.2–100.0)	0.501
D3	95.2 (95.2–100.0)	95.2 (95.2–97.6)	100.0 (95.2–100.0)	97.6 (89.3–100.0)	0.460
Timed function tests,[e] median (IQR)					
10 m walk/run, seconds	2.6 (1.8–5.0)	2.0 (1.6–2.9)	4.6 (2.5–9.7)	2.1 (1.8–3.0)	**0.017**[c,d]
Climbing four stairs, seconds	5.0 (4.0–7.0)	4.0 (3.5–6.0)	5.0 (4.5–7.5)	5.0 (4.0–7.0)	0.081

Bold type signifies β significance <0.05 level (two-tailed).

Abbreviations: BFP, body fat percentage; BMD, bone mineral density, presented by z-score; BMI, body mass index; D1, domain 1 of the MFM (standing position and transfers); D2, domain 2 of the MFM (axial and proximal motor function); D3, domain 3 of the MFM (distal motor function); FVC, forced vital capacity; IQR, interquartile range; LVEF, left ventricular ejection fraction; MRC%, the Medical Research Council grade percentage: (sum of grade scores × 100)/(number of muscles tested × 5), 12 groups of muscle were tested bilaterally, including shoulder flexors, shoulder extensors, shoulder abductors, biceps, triceps brachii, hip flexors, hip extensors, hip abductors, quadriceps, hamstrings, dorsiflexors, plantar flexors; MFM, Motor Function Measure; NIV, non-invasive ventilation; OSAS, obstructive sleep apnoea syndrome.

[a]Significant difference between groups with chronic pain and no pain.

[b]*n* (total) = 39, participants treated with steroids.

[c]Significant difference between groups with acute and chronic pain.

[d]Significant difference between groups with acute pain and no pain.

[e]*n* (total) = 39, participants who completed timed functional tests.

non-invasive ventilation. There were significant differences between pain groups (acute pain vs chronic pain) for variables including ambulation status, wheelchair dependency, Vignos grade, and 10 m walk/run test (see Table 1).

Pain characterization

Thirty-two patients reported pain during the previous 4 weeks, and 18 reported persistent or recurrent chronic pain. Of those who reported chronic pain, 13 had intermittent pain and five experienced continuous pain. All five non-ambulant and wheelchair-bound patients reported chronic pain.

Self-perceived causes of pain

Patients' self-descriptions of what they believed to be the cause of their pain were transformed into scientific language and synthesized to reveal primary themes. The most frequently stated cause of pain differed between pain groups. Exercise (n = 6) and falling (n = 5) were the most common reported causes of acute pain, while chronic pain was most often caused by muscle overuse (n = 10) and poor posture (n = 8) (Table 2). Less common causes were stretching, wearing orthoses, and muscle cramps. One patient reported chronic abdominal pain caused by the daily use of prednisone.

Frequency and episode duration of pain

Patients with chronic pain usually experienced daily and weekly pain, with most episodes lasting for hours and days (Table 3). The burden was significantly greater than patients with acute pain who generally reported pain once or several times per month, with most episodes lasting seconds to hours.

Pain location

Patients with DMD who had chronic pain showed the highest reported pain frequency across regions of the legs

TABLE 2 Self-perceived causes of pain

Self-perceived causes	Example of answers	Total (n = 32)	Pain group	
			Acute pain (n = 14)	Chronic pain (n = 18)
Overexertion	Repetitive daily activities/self-transferring/ weightlifting/manual wheelchair propulsion	12 (38)	2 (14)	10 (56)
Movement exercise	Walking/running/cycling/climbing stairs	11 (34)	6 (43)	5 (28)
Poor posture/positioning	Round back/uneven shoulders/toe walking/ hyperextended knee (bends back)/excessive lumber lordosis (swayback/belly sticks out)/prolonged wheelchair sitting	8 (25)	0	8 (44)
Trauma	Falling	5 (16)	5 (36)	0
Stretching	Daily self-stretching/passive stretching by parents or physiotherapists	3 (9)	1 (7)	2 (11)
Orthoses	Wearing ankle–foot orthoses	2 (6)	2 (14)	0
Muscle cramps	Usually occurred to calf or thigh after too much exercise or sometimes spontaneously	2 (6)	2 (14)	0
Gastrointestinal symptoms	Abdominal pain after taking prednisone	1 (3)	0	1 (6)
Do not know		6 (19)	2 (14)	4 (22)

Data are n (%).

(n = 12), lower back (n = 6), hips (n = 6), and shoulder (n = 6), of whom eight reported at least three pain sites and two reported at least five pain sites (Table 3). In addition, feet, neck, and arms were also noteworthy areas affected by chronic pain. Non-ambulant and wheelchair-bound patients were more likely to report pain in the shoulders, lower back, and hip areas, often with diffuse pain sites. Acute pain mainly affected the legs (n = 13) and feet (n = 3). Figure 1 presents the frequency of pain using localized body maps.

Pain intensity and severity

The patients' average pain intensity ratings over the previous week were graded as mild and their worst pain as moderate for patients with chronic pain. No differences were reported between pain groups. Patients with chronic pain experienced more pain within a moderate and severe range, but the differences were not statistically significant.

Pain coping, communication, and management

Pain control measures reported by young people with DMD included massage (n = 23), rest (n = 11), strategy of distraction (n = 7), heat therapy (n = 7), sleep (n = 3), and use of ice (n = 2). No patients reported using analgesics. Twenty-seven participants reported they talked to their parents about the pain they experienced, while only a small number (n = 4) actively communicated their pain complaints with clinicians during follow-up visits. Five patients reported they were forwardly asked by their physiotherapists whether they had experienced any pain, of whom three received clinical pain assessment and one received interventional management.

Patients' characteristics associated with pain variables

Table 4 demonstrates the correlations between patients' characteristics and pain measures. Older age, higher BMI, being non-ambulatory, wheelchair dependency, and the severity of scoliosis were found to be significantly associated with the presence of chronic pain. Pain intensity and pain extent were moderately correlated with age, the severity of scoliosis, and motor impairments, and, to a less degree, with BMI, ambulation status, and wheelchair dependency. No associations were detected between other characteristics and pain variables.

Psychological distress

Anxiety was present in 11 patients, with four mild, six moderate, and one severe. Depression was present in five patients, all mild. Kruskal–Wallis tests and pair comparisons showed that patients with chronic pain experienced a higher level of anxiety than those without pain (p = 0.001) and a higher level of depression compared with both the no pain (p < 0.001) and acute pain groups (p = 0.001) (Figure 2). Multiple linear regression analyses showed that chronic pain was associated with increased anxiety and depression scores on the Hospital Anxiety and Depression Scale (Table 5).

TABLE 3 Pain characteristics in patients with Duchenne muscular dystrophy reporting pain

		Group		
Pain variable	**Total (*n* = 32)**	**Acute pain (*n* = 14)**	**Chronic pain (*n* = 18)**	***p***
Pain course, *n* (%)				**<0.001**[a]
Intermittent	26 (81)	13 (93)	13 (72)	
Continuous	6 (19)	1 (7)	5 (28)	
Pain frequency, *n* (%)				**<0.001**[a]
Always, or once or several times per day	10 (31)	0	10 (56)	
Once or several times per week	9 (28)	1 (7)	8 (44)	
Once or several times per month	13 (41)	13 (93)	0	
Duration of episode, *n* (%)				<0.001[a]
Seconds	3 (9)	3 (21)	0	
Minutes	9 (28)	6 (43)	3 (17)	
Hours	13 (41)	3 (21)	10 (56)	
Days	7 (22)	2 (14)	5 (28)	
Pain extent, *n* (%)				<0.001[a]
1–2	23 (72)	13 (93)	10 (56)	
3–4	7 (22)	1 (7)	6 (33)	
≥5	2 (6)	0	2 (11)	
Pain sites,[b] *n* (%)				
Head	3 (9)	0	3 (17)	
Shoulders	6 (19)	0	6 (33)	
Arms and hands	3 (9)	0	3 (17)	
Abdomen	3 (9)	1 (7)	2 (11)	
Lower back	6 (19)	0	6 (33)	
Hips	7 (22)	1 (7)	6 (33)	
Legs	25 (78)	13 (93)	12 (67)	
Feet	7 (22)	3 (29)	4 (22)	
Pain intensity, median (IQR)				
Average pain	3.0 (2.0–4.0)	3.0 (2.0–3.2)	3.0 (2.0–4.0)	0.323
Worst pain	4.0 (3.0–5.7)	4.0 (3.0–5.0)	4.5 (3.0–6.0)	0.324
Pain severity,[c] *n* (%)				
Mild pain	21 (66)	11 (79)	10 (56)	
Moderate pain	9 (28)	3 (21)	6 (33)	
Severe pain	2 (6)	0	2 (11)	
Pain interference,[d] median (IQR)				
General activities	2.0 (1.0–3.0)	2.0 (0.7–3.0)	2.5 (1.0–4.5)	0.969
Mood	2.0 (0–3.7)	1.0 (0–2.2)	2.0 (0–5.0)	0.146
Mobility	2.0 (0–4.0)	1.5 (0.7–3.2)	2.0 (0–5.2)	0.671
School work	2.0 (0–3.0)	1.5 (0–3.0)	2.5 (0–4.2)	0.187
Relationship with others	0 (0–0.7)	0 (0–0)	0 (0–1.0)	0.058
Sleep	1.0 (0–2.0)	0 (0–1.0)	1.0 (0–4.0)	**0.013**[a]
Enjoyment of life	0 (0–2.0)	0 (0–1.0)	0.5 (0–3.0)	0.172
Average score	1.4 (0.6–2.5)	1.1 (0.4–1.7)	1.4 (0.7–3.8)	0.189

Bold type signifies β significance <0.05 level (two-tailed).

Abbreviation: IQR, interquartile range.

[a]Significant difference for the comparison of acute pain and chronic pain.

[b]One participant could have more than one site.

[c]Pain severity was rated using Numeric Pain Rating Scale scores of average pain, ratings of 0 to 3 indicating mild pain, ratings between 4 and 6 indicating moderate pain, and ratings of 7 or more indicating severe pain.

[d]Pain interference was assessed using the modified Pain Interference Scale from the Brief Pain Inventory.

HRQoL

The median overall PedsQL score was 54.2 (IQR 44.6–60.4), suggesting a relatively low quality of life in this group of patients with DMD. The subdomains with the lowest median scores, indicating the greatest impact on HRQoL, were physical functioning (50.0) and school functioning (50.0), followed by social functioning (60.0) and emotional functioning (62.5). Figure 3 presents the total score and subdomain scores of the PedsQL scale by different pain groups. Participants with chronic pain had significantly lower physical functioning and overall PedsQL scores than those without pain. Multiple linear regression analyses showed that the presence of chronic pain related to a decrease in total PedsQL scores and physical functioning scores when adjusted for age, BMI, and ambulation status, as shown in Table 6. The three groups did not differ in emotional, social, and school functioning domains.

TABLE 4 Correlations of patient characteristics with pain variables using Spearman's rank correlation coefficient

Patient characteristics	Pain condition	Pain intensity	Pain extent
Age	0.46[a]	0.51[a]	0.58[a]
Body mass index	0.45[a]	0.33[b]	0.31[b]
Ambulation level	−0.39[a]	−0.33[b]	−0.44[a]
Wheelchair dependency	0.37[b]	0.36[b]	0.36[b]
Severity of scoliosis	0.34[b]	0.40[a]	0.34[b]
Motor function measure			
Total score	−0.26	−0.41[a]	−0.42[a]
D1	−0.29	−0.45[a]	−0.46[a]

Pain condition coding: 0, no pain (n = 13); 1, acute pain (n = 14); 2, chronic pain (n = 18). Ambulation level coding: 0, non-ambulant; 1, ambulant. Wheelchair dependency coding: 0, never; 1, intermittent; 2, permanent. Severity of scoliosis coding: 0, no scoliosis; 1, mild; 2, moderate; 3, severe. Pain intensity coding: Numeric Pain Rating Scale scores of the average pain, 0–10; pain extent, pain sites, 1–5.

Abbreviation: D1, domain 1 of the Motor Function Measure (standing position and transfers).

[a]Correlation is significant at the 0.01 level.

[b]Correlation is significant at the 0.05 level.

DISCUSSION

To our knowledge, this is the first study to comprehensively discriminate between acute and chronic pain in patients with DMD and the first to identify patients' characteristics associated with chronic pain and its relation to psychosocial functioning and HRQoL from patients' perspectives. Our findings suggest that chronic pain is a frequent complaint and usually widespread in paediatric patients with DMD, highlighting the need to address pain in the routine care of affected children. Older age, higher BMI, being non-ambulatory, wheelchair dependency, and spinal deformities are important variables related to the presence of chronic pain. Furthermore, our preliminary results revealed that chronic pain seems to be significantly associated with psychological distress and reduced HRQoL in paediatric patients with DMD.

Although chronic pain is the focus of this study, we performed additional subgroup analyses by pain conditions (acute vs chronic pain) to better characterize the nature and scope of pain in young people with DMD. Most patients in our cohort reported pain (n = 32, 71%), with 14 (31%) reporting acute pain and 18 (40%) reporting chronic pain. These data were consistent with previous studies of paediatric cohorts with dystrophinopathies (DMD and Becker muscular dystrophy), with around 54% to 70% having self-reported pain[12,26] and 41% chronic pain.[26] The study conducted by Lager et al.[26] was the only one investigating the prevalence, characteristics, and impact of chronic pain in adolescents with DMD. However, it did not address potential causes and patients' characteristics associated with chronic pain in this group. It is noteworthy that our estimate of chronic pain was much lower than those reported in the adult population with dystrophinopathies (65–67%).[7,10,27] This is to be anticipated, because patients with DMD experience pain as an expected

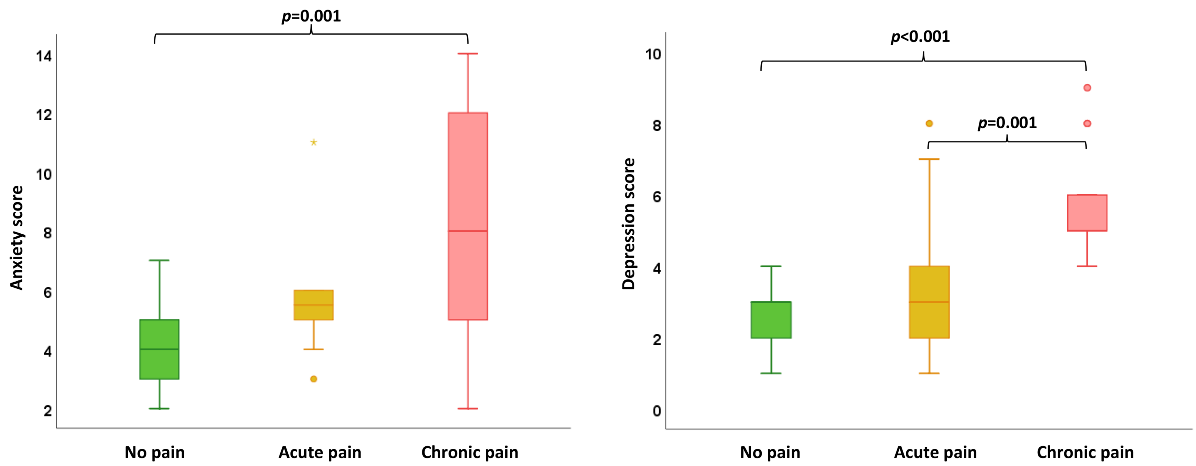

FIGURE 2 Anxiety and depression scores by pain groups. Anxiety and depression were rated using the Hospital Anxiety and Depression Scale

TABLE 5 Multiple linear regression model including the Hospital Anxiety and Depression Scale with independent variables: chronic pain, age, body mass index, and ambulation status

Dependent variables	Independent variables	R	p	R^2	Adjusted R^2	β	95% CI	SE	Standardized β	p
HADS anxiety	Chronic pain	0.545	**0.006**	0.297	0.226	1.573	0.260–2.886	0.650	0.382	**0.020**
	Age					0.067	−0.592 to 0.726	0.326	0.037	0.839
	Body mass index					0.193	−0.242 to 0.628	0.215	0.138	0.376
	Ambulation status					−1.293	−5.226 to 2.639	1.946	−0.120	0.510
HADS depression	Chronic pain	0.636	**<0.001**	0.405	0.346	1.637	0.887–2.388	0.371	0.639	**<0.001**
	Age					−0.076	−0.453 to 0.301	0.186	−0.068	0.686
	Body mass index					−0.060	−0.309 to 0.189	0.123	−0.069	0.628
	Ambulation status					−0.862	−3.111 to 1.386	1.112	−0.129	0.443

Bold type signifies β significance <0.05 level (two-tailed).

Abbreviations: β, unstandardized coefficients; Standardized β, standardized coefficients; CI, confidence interval; HADS, Hospital Anxiety and Depression Scale; R, the correlation coefficient between the predicted values and the observed values of Y; R^2, the coefficient of determination; adjusted R^2, a modified version of R^2 that has been adjusted for the number of predictors in the model; SE, standard error.

result of muscle weakness and skeletal abnormalities that progress rapidly with age.

Findings from subgroup analyses indicated significant differences in the nature and extent of acute and chronic pain in paediatric patients with DMD. Acute pain mainly affects the legs, whereas chronic pain often involves multiple locations across the legs, lower back, shoulders, and hips. Furthermore, the findings of pain frequency and duration of episodes suggested that chronic pain may place a significantly greater burden on the paediatric DMD population. The two groups did not differ in the mean value of pain intensity (mild on average and moderate in worst pain). However, a higher proportion of patients with chronic pain described their pain as moderate and severe than those with acute pain.

Although further confirmation is needed, the findings from patients' self-perceived causes of pain were indicative. It is important to note that exercise and falling are frequent causes of acute pain in ambulatory patients with DMD. Muscle weakness, altered gait pattern (e.g. toe walking), and poor balance control can lead to repeated falls in the early stages of DMD, resulting in painful soft-tissue injuries that may last for hours or days. No patients reported bone fractures in this study. Nevertheless, previous studies have suggested that the fracture rate is high in patients with DMD and is most often precipitated by falls.[28] Bone fractures of limbs are likely to cause severe pain lasting for weeks and months and impact the subsequent mobility of the patient.

Muscle pain, particularly after physical activity, is a major complaint in male children with DMD.[29,30] In our study, muscle pain after movement exercise was reported by a medium proportion of patients. Certain movement forms included prolonged walking, climbing stairs, cycling, and running. As described by the patients, the pain occurred mainly in the calf and thigh, lasted a few seconds or minutes, and recovered after rest or massage at most. However, the pain usually recurred when the same form of exercise was performed, and a certain dose was reached. The problem of exercise training and contraction-induced muscle damage remains a large clinical concern.[29] Excessive or prolonged muscle pain during or after physical activities may be associated with eccentric exercise-induced muscle damage[29,31] or overexertion.[32]

For people diagnosed with muscular dystrophies, pain may transition from acute to chronic owing to persistent or recurrent noxious stimuli. There are critical differences in the disease burden, impacts, and management of chronic pain compared with acute, transient pain conditions.[33,34] Chronic pain in patients with DMD is often multifactorial, including severe contractures, muscle overuse, poor posture, spinal deformity, and fractures.[8,32] In our study, the participants were mainly independent walkers (n = 40), where severe contractures, vertebral fractures, and severe scoliosis were rarely reported. The primary sources of chronic musculoskeletal pain are more likely to be from muscle fatigue and overuse symptoms (muscle sprain or strain) and mechanical problems or pathokinesiological maladaptation

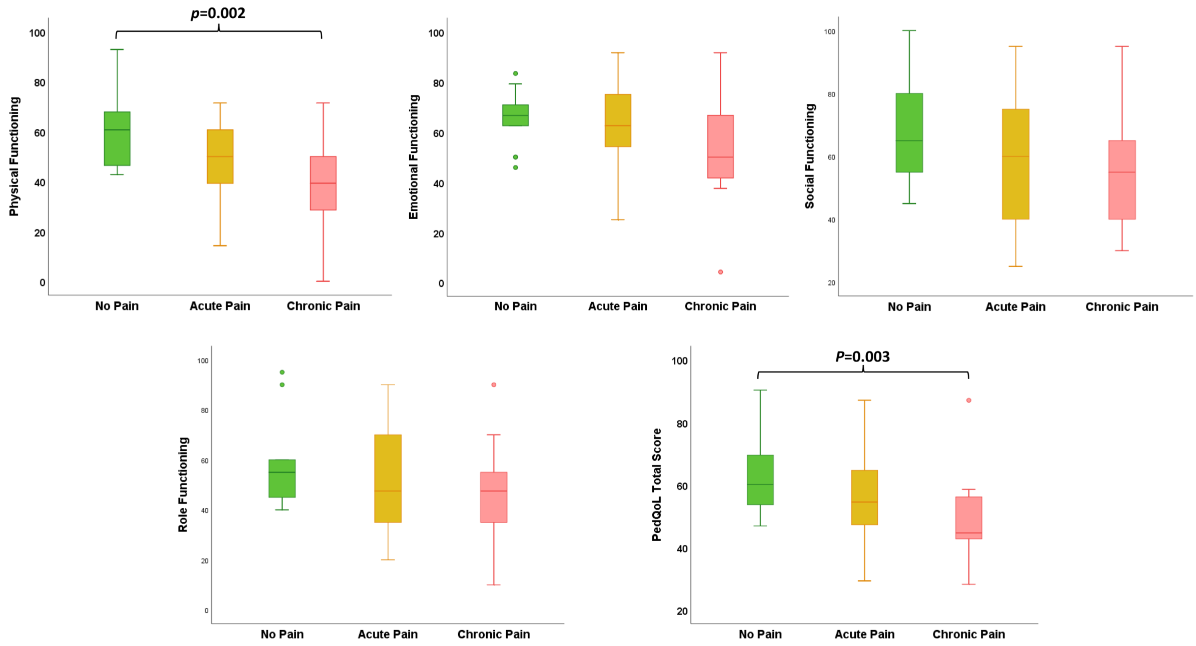

FIGURE 3 Total and subdomain scores of the Pediatric Quality of Life Inventory in patients with Duchenne muscular dystrophy by pain group

related to poor posture. Altered gait pattern in individuals with DMD has been proposed as a potential reason for the high frequency of pain in the back and legs, common features including excessive anterior pelvic tilt, knee hyperextension in the loading response phase, and an over-plantarflexed foot.[35] Pathokinesiological adaptation of prolonged daily toe walking is also thought to be a reason for occasional localized calf pain.[32]

To the best of our knowledge, no previous studies have investigated the cause of pain in patients with DMD. The phenotype of DMD-related chronic pain is largely unknown and may involve several different mechanisms. The predominant pain involvement in the legs, hips, back, and shoulders suggests a musculoskeletal origin. Thus, chronic nociceptive pain seems to be the most common type of pain described and is presumed to arise because of profound changes in the structure and function of tissues within the musculoskeletal system. Given the widespread and prolonged nociceptive input associated with DMD, it seems likely that central sensitization and a mixed nociceptive/nociplastic pain phenotype may develop, at least in a subgroup of people. However, this has not yet been explored in people with DMD. Similarly, clinical evidence seems to be limited for neuropathic pain in individuals with DMD. An in-depth understanding of the mechanisms and phenotypes of pain in DMD should be the subject of future studies.

Pain in patients with DMD is considered an expected consequence of progressive musculoskeletal impairments. As anticipated, older age, being non-ambulatory, wheelchair dependency, and the severity of spinal deformities were found to be significantly associated with the presence of chronic pain. While losing the ability to walk and being confined to a wheelchair as the diso advances is inevitable for people with DMD, chronic pain is an unfortunate fact of daily life for many wheelchair users. Our findings suggested that patients with DMD with increased wheelchair dependency are more likely to experience widespread chronic pain across the neck, back, and shoulder regions, possibly as a result of repetitive loading (induced by activities of daily living, such as transferring and weight relief tasks and wheelchair propulsion) and poor posture/positioning (linked to prolonged sitting in an unergonomic wheelchair and progressive scoliosis).[32] BMI was also found to correlate with the presence of chronic pain. Weight gain in DMD is often seen during the transition to the non-ambulatory stage and may be ascribed to a decrease in physical activity and an increase in corticosteroid use and caloric intake.[36] No participants were identified as obese in this study. However, it should be noted that obesity is not uncommon in patients with DMD. The condition may hinder movement, accentuate skeletal deformity, and increase general discomfort during wheelchair use, which may potentially contribute to the occurrence of pain.

Given the progressive and disabling nature of the males with DMD are at higher risk of affective disorders than their typically developing peers or other paediatric patient groups with physical disabilities.[37] Anxiety and depression were present in 24% and 11% of our cohort respectively. The rates were high in light of estimates in the general paediatric population (anxiety 5.2%, depression 1.3%)[38] but relatively low compared with those found in previous studies, including adolescent and adult cohorts with DMD (anxiety 24–27%, depression 19–29%).[7,39,40] Chronic pain is commonly comorbid with a depressive or anxiety disorder. The co-occurrence rate of pain and affective disorders was 29.6% in an adult cohort with DMD.[7] In line with findings in other paediatric patient groups,[41] our multiple linear regression analyses demonstrated that the presence of chronic pain was significantly associated with increased anxiety and depressive

TABLE 6 Multiple linear regression model including Pediatric Quality of Life Inventory with independent variables: chronic pain, age, body mass index, and ambulation status

Dependent variables	Independent variables	R	P	R^2	Adjusted R^2	β	95% CI	SE	Standardized β	p
Physical functioning	Chronic pain	0.687	**<0.001**	0.473	0.420	−7.357	−13.620 to −1.094	3.099	−0.324	**0.022**
	Age					−1.668	−4.810 to 1.475	1.555	−0.168	0.290
	Body mass index					0.152	−1.923 to 2.228	1.027	0.020	0.883
	Ambulation status					21.873	3.117–40.629	9.280	0.368	**0.023**
Overall HRQoL	Chronic pain	0.514	**0.014**	0.264	0.190	−5.852	−11.363 to −0.342	2.727	−0.346	**0.038**
	Age					1.363	−1.402 to 4.128	1.368	0.184	0.325
	Body mass index					−0.816	−2.642 to 1.010	0.903	−0.142	0.372
	Ambulation status					12.457	−4.046 to 28.961	8.166	0.281	0.135

Bold type signifies beta significance <0.05 level (two-tailed).

Abbreviations: β, unstandardized coefficients; Standardized β, standardized coefficients; CI, confidence interval; HRQoL, health-related quality of life; R, the correlation coefficient between the predicted values and the observed values of Y; R^2, the coefficient of determination; adjusted R^2, a modified version of R^2 that has been adjusted for the number of predictors in the model; SE, standard error.

symptoms in young people with DMD, suggesting that chronic pain may interact with psychological distress in paediatric patients with DMD. It has been proposed that chronic pain and depression most probably have a bidirectional association and exacerbate one another.[42,43] Thus, addressing psychological distress in rehabilitation interventions may be beneficial for patients with DMD and could potentially reduce both pain and psychological challenges.

In general, HRQoL in patients with muscular dystrophies is low and associated with disease progression.[44] The present study found similar results that HRQoL was substantially impaired across various physical and psychosocial functioning domains in male children with DMD. Patients with DMD who have chronic pain seem to have poorer HRQoL, which is in line with findings in the broader pain literature[45] and studies of pain in other types of neuromuscular disorder.[5,6] The presence of chronic pain was found to be significantly associated with increased deficits in physical functioning and overall HRQoL. However, in a study that included an adult cohort with DMD, it was stated that chronic pain was only associated with the domain of physical health and did not significantly affect the overall HRQoL, as assessed by using the World Health Organization Quality of Life Scale - Brief Version.[7] Nonetheless, psychometric studies using Rasch analysis have shown that the PedsQL total score used by the present study may not be optimal as a measure of HRQoL in the DMD group;[46] thus, future investigation using a more scientifically robust measure would be beneficial to confirm the association between chronic pain and overall HRQoL in this patient group.

Findings from this study highlight that chronic pain is a common and significant problem in male children with DMD that interferes with daily life activities across various domains and is significantly associated with psychological distress and reduced HRQoL. Despite this, pain in these people is under-assessed and undertreated in clinical practice. Owing to a lack of knowledge of the nature and impact of pain in DMD, health professionals may not view pain as one of the most disabling aspects in patients with DMD and forwardly assess their pain complaints. On the other hand, from the patients' perspectives, pain complaints are usually kept within the family, and most male children in our cohort reported they were reluctant to actively communicate about their pain with health professionals. Poor pain communication in children can lead to underestimation of pain; meanwhile, accurate pain localization and intensity reporting are critical to guide effective pain management. We recommend that, during all routine clinic visits, health professionals ask the child/adolescent and caregivers about pain complaints and take a comprehensive pain history. The body map has been found to be a valid and reliable tool to elicit information about the location of pain symptoms from children.[47] School-age children, particularly by 8 years of age, are relatively accurate in communicating their pain and can very reliably describe the location of the pain using a body map. To address and treat pain early on is of great significance in DMD. Otherwise, the pain may transition from acute to chronic, which can exacerbate a child's HRQoL by interfering with physical activities, sleep, mood, school attendance, and social participation.

Considering the progressive and disabling nature of DMD, the multifactorial mechanisms of chronic pain, and its significant impact on various life domains of the person affected, a plan of care for pain management and prevention needs to be developed in collaboration with the multidisciplinary team. In light of the current practice guidelines for DMD, comprehensive interventions should include physiotherapy, postural correction, orthotic intervention and splinting, wheelchair and bed enhancements that allow independent weight shift, position change, and pressure relief.[3] Apart from pain control, such interventions should aim at preventing trauma, muscle damage, movement-induced pain, overexertion, and poor posture. Interventions should also address the disabilities related to pain which include limitations in physical activities, impairments in school functioning and social participation, and the potential for emotional distress and depressive symptoms. Such an approach may both reduce pain and help to improve the various domains of life experience that can be affected by pain, which ultimately contribute to improving the quality of life in this population.

The following limitations of our study should be considered when interpreting the results. First, the cross-sectional design does not allow conclusions about which directions the relations between pain and anxiety, depression, and HRQoL go. Second, patients were recruited from a high-resource healthcare setting and were all included in a multidisciplinary care programme. This may affect the study's external validity to the general paediatric population with DMD. Furthermore, the study sample was relatively young, which may also threaten the generalizability of our findings. Third, given the limited number of included patients, caution must be taken about some of the conclusions. Results generated from multiple linear regression analyses may be susceptible to change if more confounding variables are controlled. In our analyses, only age, BMI, and ambulatory status were adjusted owing to sample size restriction. Finally, we cannot rule out recall bias and selection bias as it was a retrospective study based on subjective patient reports.

CONCLUSION

Our findings contribute to the limited literature on chronic pain in paediatric neuromuscular disorders. Chronic pain in paediatric DMD is frequent and widespread, and it seems to differ in aetiology, scope, and nature compared with transient, acute pain. The most common reported self-perceived causes of chronic pain are related to muscle overuse and poor posture. Average pain intensity is mild, with legs, low back, hips, and shoulders most affected. Older age, higher BMI, being non-ambulatory, wheelchair dependency, and severity of scoliosis are important variables related to the presence of chronic pain. Furthermore, chronic pain is significantly associated with psychological distress and reduced HRQoL in paediatric patients with DMD. Population-based prospective cohort studies are needed to determine the causal relationship between patient characteristics, psychological distress, and chronic pain, and to confirm the extent to which pain contributes to reduced HRQoL in paediatric patients with DMD. This may help facilitate the development of targeted therapeutic strategies to reduce pain and improve HRQoL.

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CONFLICT OF INTEREST

The authors have stated that they had no interests that might be perceived as posing a conflict or bias.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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